

CS 230 : Discrete Computational Structures

Fall Semester, 2023

ASSIGNMENT #8

Due Date: Thursday, November 2

Suggested Reading: Rosen Sections 5.3; Lehman et al. Chapter 7

For the problems below, explain your answers and show your reasoning.

1. [12 Pts] Let S defined recursively by (1) $4 \in S$ and (2) if $s \in S$ and $t \in S$, then $st \in S$. Let $A = \{4^i \mid i \in \mathbf{Z}^+\}$. Prove that
 - (a) [6 Pts] $A \subseteq S$ by mathematical induction.
 - (b) [6 Pts] $S \subseteq A$ by structural induction.
2. [5 Pts] Give an inductive definition of the set of palindromes over the alphabet $\{a, b, c\}$. You do not need to prove that your construction is correct. *Note:* a, b, c, aa, cc, aba are all palindromes.
3. [5 Pts] Define the set $S = \{2^k 3^m \mid k, m \in \mathbf{Z}^+\}$ inductively. You do not need to prove that your construction is correct.
4. [8 Pts] Given the inductive definition of full binary trees (FBTs), define $n(T)$, the number of vertices in tree T , and $\ell(T)$, the number of leaves in tree T , inductively. Then, use structural induction to prove that for all FBTs T , $n(T) = 2\ell(T) - 1$.
5. [15 Pts] Let $L = \{(a, b) \mid a, b \in \mathbf{Z}, (a - b) \bmod 4 = 0\}$. We want to program a robot that can get to each point $(x, y) \in L$ starting at $(0, 0)$.
 - (a) [5 Pts] Give an inductive definition of L . This will describe the steps you want the robot to take to get to points in L starting at $(0, 0)$. Let L' be the set obtained by your inductive definition.
 - (b) [5 Pts] Prove inductively that $L' \subseteq L$, i.e., every point that the robot can get to is in L .
 - (c) [5 Pts] **Extra Credit** Prove that $L \subseteq L'$, i.e., the robot can get to every point in L . To prove this, you need to give the path the robot would take to get to every point in L from $(0, 0)$, following the steps defined by your inductive rules.

For more practice, you are encouraged to work on other problems in Rosen Sections 5.3 and in LLM Chapter 7.